Busines1.s Plan: FreshBake Bread Factory

FreshBake Bread Factory is a small-scale bakery that produces and sells a variety of freshly baked bread and pastries to local retailers and individual customers. The factory aims to offer high-quality, affordable baked goods while maintaining efficient production through the combination of fixed and variable factors.

Factors of Production: Fixed Factors (do not change with output in the short run):

Factory building – Rented space for production

Ovens and baking equipment – Industrial ovens, mixers, and baking trays

Office equipment– Computers, furniture, and phones for admin staff

Variable Factors (change with level of output):

Raw materials – Flour, sugar, yeast, eggs, etc.

Labour– Bakers, packers, cleaners, delivery drivers (can be increased or reduced depending on demand)

Utilities – Electricity and water usage increase as production increases

Production Process

1. Raw materials are delivered daily.

2. Bakers mix ingredients and prepare bread.

3. Bread is baked using industrial ovens.

4. Goods are cooled, packaged, and distributed.

Application of the Law of Diminishing Returns

Law of Diminishing Returns states that if one factor of production (e.g. labour) is increased while other factors (e.g. ovens or factory space) are held constant, there comes a point where the marginal output (extra output from one more unit of input) starts to DIMINISH.

As more and more units of a variable factor (e.g., labor) are added to fixed factors (e.g., land or capital), the additional output (marginal product) from each additional unit of the variable factor will eventually diminish.

Example in FreshBake Factory

Suppose FreshBake has 2 industrial ovens (fixed factor).

Initially, with 2 bakers, the ovens are used efficiently.

Adding a 3rd or 4th baker may increase total output as they share the workload.

However, after a point (say 6 bakers), the kitchen becomes crowded, bakers wait to use ovens, and coordination problems occur.

Result: Each additional baker adds less and less to total output, or may even reduce efficiency.

Conclusion: While hiring more workers can increase production initially, without expanding fixed factors like oven capacity or space, productivity per worker will eventually fall — demonstrating the law of diminishing returns.

Strategies to Manage Diminishing Returns

Invest in additional ovens or expand workspace when increasing labour.

Introduce shifts to spread production over longer hours rather than crowding workers.

Improve workflow through automation or training.

Conclusion

FreshBake's success depends on wisely managing its fixed and variable inputs. By understanding and planning for the law of diminishing returns, the business can maintain efficient production and sustain profitability.